Multi-Agent Geopolitical Intelligence Assistant

Technical Specification - APEX:E3 - Oxford Group Design Practical - Hilary / Trinity Term 2026

1. Overview

Build a three-agent pipeline that generates a daily geopolitical intelligence report. Runs locally, single command, no cloud. Prioritise: (1) Report quality, (2) Structural clarity, (3) Reliability, (4) UI.

Outputs: HTML report, JSON structured file, MCP communication logs, pipeline timestamps.

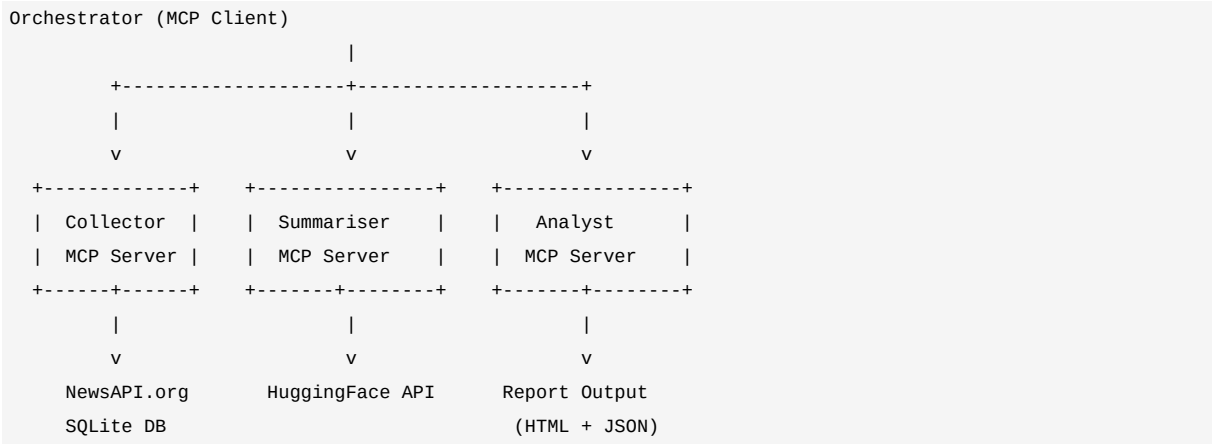
2. Deliverables

HTML Report (readable in <10 minutes): report metadata (date, timestamp, sources), 5-10 events from previous 24h (region, key actors, 3-5 sentence summary, sentiment score, relevance score, impact assessment across 4 categories, source links), and 2-3 Themes of the Day.

JSON Output: all event data, scores, source metadata, agent logs, pipeline timestamps.

3. Architecture: Orchestrator + MCP Tool Servers

One Orchestrator (MCP client) calls three Agent Servers (MCP servers) sequentially. Agents never call each other - the orchestrator owns sequencing, logging, and error handling.



Agent	Tool Exposed	Role
Collector	collect_articles()	Query NewsAPI, deduplicate, store S
Summariser	summarise_batch()	Cluster articles, extract region/ac
Analyst	analyse_events()	Score relevance, impact assessment

Pipeline: articles = collector.call_tool(...) -> events = summariser.call_tool(...) -> report = analyst.call_tool(...) -> write outputs.

4. Impact Assessment

Each event scored across 4 categories. Each gets Low/Medium/High severity + one-sentence justification. If no impact: "No material impact identified." Honest nulls > hallucinated outputs.

Category	Covers
Economic	Trade/sanctions, supply chains, com
Financial Markets *	FX, equity sector exposure, soverei
Security	Military escalation, terrorism, cyb
Diplomatic	Alliance shifts, IO responses, sanc

* Financial Markets must be rule-based. Config file mapping: Region -> sectors, Event type -> asset class, Commodity -> sectors. LLM writes narrative only - classification uses rules.

5. Stack

Component	Choice
Language	Python
Agent comms	MCP Python SDK (mandatory)
Database	SQLite
Models	HuggingFace: bart-large-cnn, twitte
Orchestration	Docker Compose
Data	NewsAPI.org (primary), GDELT REST A

DB tables: articles, events, impact_scores, reports, agent_messages, pipeline_runs

6. User, Docs & Ethics

Persona: mid-level analyst at a financial services / policy org needing a concise morning briefing. Not needed: real-time alerts, social media, price forecasting.

Docs: README (install-from-cold), architecture diagram, evaluation report (3-4pp).

Ethics: discuss misinformation risk, Western source bias, liability, intended vs unintended use.

7. Success Criteria

Pass: pipeline runs end-to-end, orchestrator calls 3 agents via MCP, HTML readable, JSON valid, docs usable.

High quality: error handling, API failure recovery, clean schema, clear reasoning, useful writing, honest evaluation.

Stretch (only if ahead): Streamlit UI or Reviewer/Critic agent. A well-executed smaller system beats an ambitious one that partially works.